

# Where is the lesion? (I)

## Symptoms and signs in Neurology

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# Neurology

- ◉ **Neurology:** branch of medicine dealing with diagnosis & treatment of diseases involving brain, spinal cord, spinal roots, peripheral nerves &/or muscles
- ◉ **Neurologist:** physician specializing in neurology; trained to investigate, diagnose & treat neurological disorders

# Approach to patients with neurological complaints

- ◉ Diagnosis from **history** alone in many cases (80-90%)
- ◉ Focused on 3 key questions:
  - > **Where is the lesion?**
  - > What is the underlying **pathology/aetiology?**
  - > Is it neurological or **functional?**

# History taking

- History will direct our neurological examination
- Examination can confirm history & support diagnosis
- Relative or eyewitness
- History structure: C/O, HPI (onset, duration, course, recurrence, associated symptoms), PMH (risk factors), drug & allergy, family, social, O&G, review of systems

# Good history taking

- ◉ Both open & closed questions
- ◉ Chronicity & associated features
- ◉ System-specific questions
- ◉ Not the same as a dictation/translation
- ◉ Avoid irrelevant information/too much details
- ◉ Minimum of important 'negatives'
- ◉ Ideas, concerns, expectations
- ◉ Reflecting experience & knowledge

# Physical examination

- Elicit signs (according to history & working diagnosis); new vs. pre-existing
- General examination (stay focused)
- Neurological examination (screening vs. focused)
- Examinations of other systems
- **Interpretation:** normal vs. abnormal; anatomical & aetiological diagnosis

# History & physical examination

- Do not rely on just one isolated symptom or sign, especially if it goes against the rest of history or examination findings
- 'Hard' physical signs
  - > Wasting
  - > Fasciculations
  - > Limb reflexes, Babinski's sign
  - > Nystagmus

# History & physical examination

- ◎ 'Hard' physical signs
  - > Dysconjugate gaze
  - > Gag reflex
  - > Corneal & pupillary reflexes
  - > Cog-wheel rigidity
- ◎ 'Soft' physical signs
  - > Power
  - > Unsteadiness
  - > Sensory deficits



# Central & peripheral nervous system

- ◉ Central nervous system = brain & spinal cord
- ◉ Peripheral nervous system = 12 pairs of cranial nerves & 31 pairs of peripheral nerves

# Anatomical structures

- ◉ Brain
  - > Cerebral hemispheres (left, right)
  - > Brain stem (midbrain, pons, medulla)
  - > Cerebellum
- ◉ Spinal cord & spinal roots
- ◉ Cervical, brachial, lumbar, sacral, coccygeal plexuses
- ◉ Cranial & peripheral nerves (motor, sensory, mixed)
- ◉ Neuromuscular junctions
- ◉ Muscles

# Brain

- ◉ Hemispheres: cortex, white matter, deep nuclei, ventricles
- ◉ Brain stem: mid-brain, pons, medulla
- ◉ Cerebellum
- ◉ 2% of body weight; 15% of cardiac output; 20% of oxygen intake

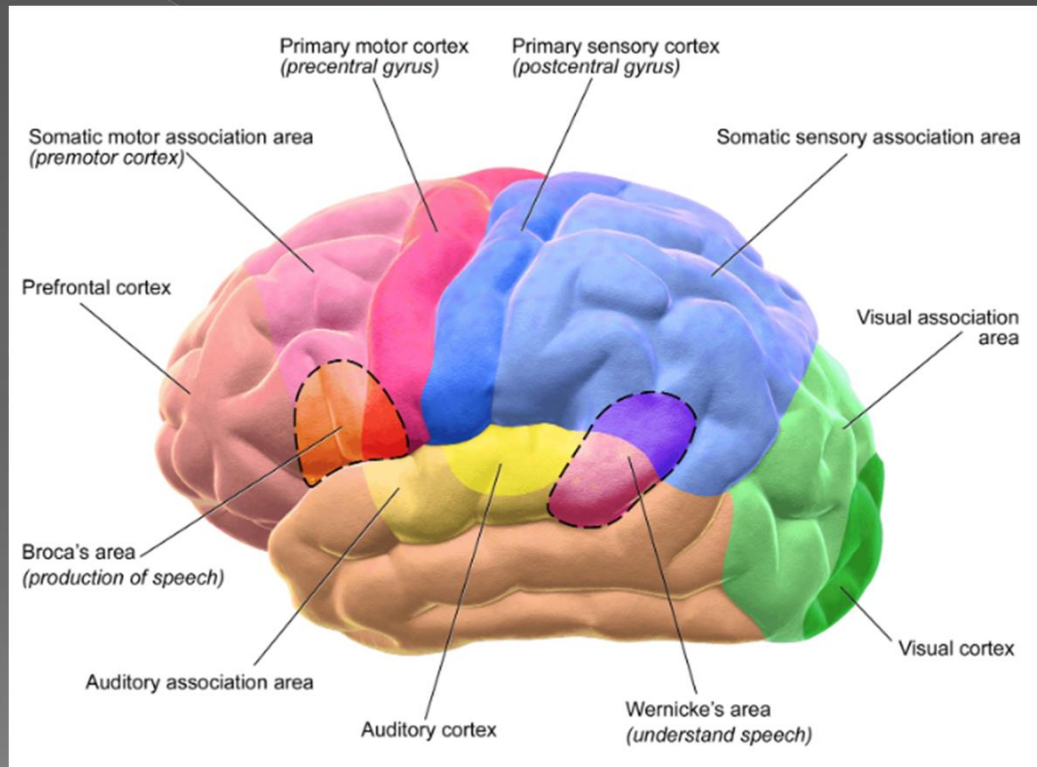
# Grey matter

- ◉ Neurone cell bodies
- ◉ **Cortex** folded into bulges (gyri) with deep furrows or fissures (sulci)
- ◉ **Deep nuclei (basal ganglia & thalami)**
- ◉ Information processing
- ◉ Right & left hemispheres
- ◉ About 2/3 of hemispheres

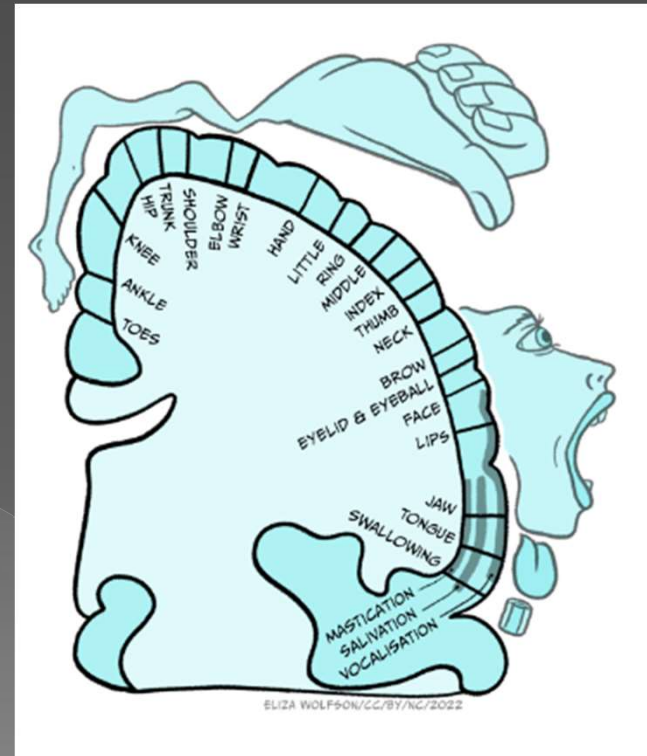
# Cerebral cortex

- ◉ Most highly developed part
- ◉ Most recent structure in evolution
- ◉ Six distinct layers: I to VI
- ◉ Sensing & interpreting inputs from various sources
- ◉ Cognitive functions
- ◉ Motor control & execution
- ◉ Lobes with specific functions

# Primary, secondary & tertiary cortices



<https://human-memory.net/about/>



<https://openpress.sussex.ac.uk/introductiontobiologicalpsychology/chapter/the-motor-system/>

# Limbic system

- ◉ Subcallosal area, cingulate gyrus, parahippocampus, uncus, hippocampal formation, nucleus accumbens, hypothalamus, mammillary bodies, amygdala
- ◉ Behavioural & emotional responses, especially needed for survival (feeding, reproduction, caring for our young, & fight or flight responses)

# Higher Cortical Functions

- Cerebral cortex & ascending reticular activating system: consciousness & attention
- Frontal: problem-solving, self-motivation, planning, mental tracking, abstract thinking, general motor control, language expression
- Limbic: memory & emotion



# Higher Cortical Functions

- ◉ **Parietal:** perception & integration of sensory data (stereognosis, graphesthesia, attention & recognition to simultaneous stimuli), space orientation, complex movement, recognition of self & the world
- ◉ **Temporal:** auditory information, language reception, encoding of memory, affect/emotions
- ◉ **Occipital:** vision & related information

# Functional systems

- ◉ **Dominant** hemisphere for language: left in 99% Rt-handed, 50% Lt-handed
- ◉ **Non-dominant** hemisphere for spatial awareness & visual processing
- ◉ **Pyramidal system**: voluntary control of movement
- ◉ **Extrapyramidal system**: involuntary control of movement & coordination

# Functional systems

- ◉ Cerebellar system: balance, coordination
- ◉ Autonomic system: HR, BP, respiration, GI, sweating, pupils, urination, sexual
- ◉ Spinothalamic system: crude touch, pain, temp
- ◉ Dorsal column system: fine touch, JPS, vibration

# Higher mental functions

- ◉ Ability to provide a coherent history
- ◉ Speech & language
- ◉ Glasgow Coma Scale (eye, verbal, motor)
- ◉ Mini-Mental State Examination (MMSE):  
orientation to time & place, immediate &  
short-term recall, serial 7, repetition, naming of  
2 objects, 3-step command, read & obey  
command, writing a sentence, copying of a  
drawing; Montreal Cognitive Assessment  
(MoCA)

# Mini-mental state examination

## Mini-Mental State Examination (MMSE)\*

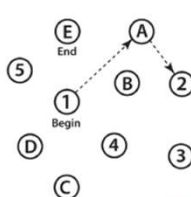
| Domain and Test                                                               | Score (maximum)   |
|-------------------------------------------------------------------------------|-------------------|
| <i>Orientation</i>                                                            |                   |
| Ask for year, season, month, date, day                                        | _____ (5)         |
| Ask for country, town, district, hospital, floor                              | _____ (5)         |
| <i>Registration</i>                                                           |                   |
| Name three objects for patient to repeat (e.g. apple, spectacles, chopsticks) | _____ (3)         |
| <i>Attention and Calculation</i>                                              |                   |
| Serial 7 subtraction from 100                                                 | _____ (5)         |
| <i>Recall</i>                                                                 |                   |
| Ask for the three objects learnt in <i>Registration</i>                       | _____ (3)         |
| <i>Language</i>                                                               |                   |
| Name a pencil and watch                                                       | _____ (2)         |
| Repeat 'No ifs, ands, or buts'                                                | _____ (1)         |
| Follow a 3-stage command                                                      | _____ (3)         |
| Read and obey a written command                                               | _____ (1)         |
| Write a complete sentence                                                     | _____ (1)         |
| <i>Copy</i>                                                                   |                   |
| Copy a pair of intersecting pentagons                                         | _____ (1)         |
| <b>Total MMSE Score</b>                                                       | <b>_____ (30)</b> |
| *>27 = normal and <24 = dementia                                              |                   |

# Montreal cognitive assessment

**MONTREAL COGNITIVE ASSESSMENT (MOCA®)**  
Version 8.1 English

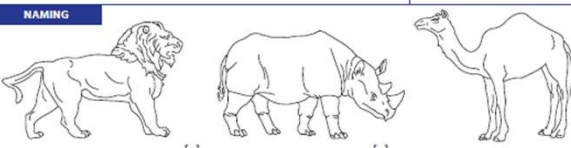
Name: \_\_\_\_\_ Education: \_\_\_\_\_ Sex: \_\_\_\_\_ Date of birth: \_\_\_\_\_ DATE: \_\_\_\_\_

**VISUOSPATIAL/EXECUTIVE**

Copy cube  [ ]

Draw CLOCK (Ten past eleven) (3 points) [ ] [ ] [ ]

**NAMING**

 [ ] [ ] [ ]

**MEMORY**

Read list of words; subject must repeat them. Do 2 trials, even if 1st trial is successful. Do a recall after 5 minutes.

|                       | FACE | VELVET | CHURCH | DAISY | RED |
|-----------------------|------|--------|--------|-------|-----|
| 1 <sup>st</sup> TRIAL |      |        |        |       |     |
| 2 <sup>nd</sup> TRIAL |      |        |        |       |     |

**ATTENTION**

Read list of digits (1 digit/sec). Subject has to repeat them in the forward order: [ ] 2 1 8 5 4

Subject has to repeat them in the backward order: [ ] 7 4 2

Read list of letters. The subject must tap with his hand at each letter A. No points if ≥ 2 errors

[ ] F B A C M N A A J K L B A F A K D E A A A J A M O F A A B

Serial 7 subtraction starting at 100. [ ] 93 [ ] 86 [ ] 79 [ ] 72 [ ] 65

4 or 5 correct subtractions: 3 pts. 2 or 3 correct: 2 pts. 1 correct: 1 pt. 0 correct: 0

**LANGUAGE**

Repeat: I only know that John is the one to help today. [ ]

The cat always hid under the couch when dogs were in the room. [ ]

Fluency: Name maximum number of words in one minute that begin with the letter F. [ ] (N ≥ 11 words)

**ABSTRACTION**

Similarity between e.g. banana - orange = fruit [ ] train - bicycle [ ] watch - ruler

**DELAYED RECALL**

Memory Index Score (MIS)

|    | FACE | VELVET | CHURCH | DAISY | RED |
|----|------|--------|--------|-------|-----|
| X1 |      |        |        |       |     |
| X2 |      |        |        |       |     |
| X3 |      |        |        |       |     |

MIS = \_\_\_\_/15

**ORIENTATION**

[ ] Date [ ] Month [ ] Year [ ] Day [ ] Place [ ] City

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Administered by: \_\_\_\_\_ MIS: \_\_\_\_/15 (Normal ≥ 26/30)

Training and Certification are required to ensure accuracy Add 1 point if ≤ 12 yr old

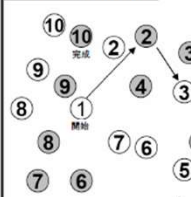
TOTAL \_\_\_\_/30

**Montreal Cognitive Assessment Hong Kong Version (HK-MoCA)**

姓名: \_\_\_\_\_ 教育年數: \_\_\_\_\_ 性別/年齡: \_\_\_\_\_ 日期: \_\_\_\_\_

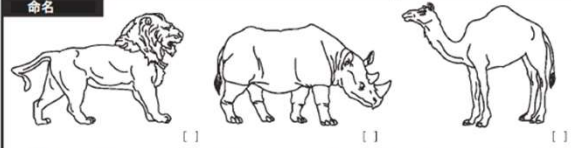
蒙特利爾認知評估香港版

**視覺空間/執行功能**

複製圖形  [ ]

畫時鐘 (十一點十分) (3分) [ ] [ ] [ ]

**命名**

 [ ] [ ] [ ]

**記憶**

讀出詞語再由病者重複

|       | 面孔 | 絲絨 | 教堂 | 薔薇 | 紅色 |
|-------|----|----|----|----|----|
| 第一次嘗試 |    |    |    |    |    |
| 第二次嘗試 |    |    |    |    |    |

5分鐘後回憶

**專注**

讀出數字 (每秒一個)

病者須把數字向前重複 [ ] 2 1 8 5 4

病者須把數字向後重複 [ ] 7 4 2

讀出數字: 當數字 "1" 出現時病者必須用手敲打桌面 (如 ≥ 2 錯誤便不給予分數)

[ ] 5 2 1 3 7 4 1 1 8 0 6 2 1 5 1 7 4 5 1 1 1 4 1 7 0 5 1 1 2

由 100 開始連續 7 減算 [ ] 93 [ ] 86 [ ] 79 [ ] 72 [ ] 65

4 或 5 個正確減算得 3 分, 2 或 3 個正確得 2 分, 1 個正確得 1 分, 沒有正確得 0 分

**語言**

重複: 姨丈買魚腸 [ ]

西施十四歲 [ ]

流暢: 一分鐘內能說出的動物名稱的數目 [ ] (N ≥ 11 個名稱)

**抽象**

相似點: 例如: 香蕉 - 橙 = 生果 [ ] 火車 - 單車 [ ] 手錶 - 間尺

**延遲記憶**

|       | 面孔 | 絲絨 | 教堂 | 薔薇 | 紅色 |
|-------|----|----|----|----|----|
| 第一次嘗試 |    |    |    |    |    |
| 第二次嘗試 |    |    |    |    |    |

5分鐘後回憶

每日提示 (見下表)

多項選擇 (見下表)

**定向**

[ ] 日 [ ] 月 [ ] 年 [ ] 星期 [ ] 地點 [ ] 地區

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Hong Kong Version updated 26 Oct 2018  
Adapted by Hong & Lau MoCA V  
http://www.mocatest.org

區選記憶測試表

| 題目提示 | 多項選擇    |
|------|---------|
| 面孔   | 身體的一部分  |
| 絲絨   | 絲絨品的一部分 |
| 教堂   | 建築物的一部分 |
| 薔薇   | 花的一部分   |
| 紅色   | 顏色      |

總分 \_\_\_\_/30

百分比分數 (見後頁)

□ > 16% □ ≤ 16% □ ≤ 7% □ ≤ 2%

# Localization: cerebral cortex

- Expressive aphasia, impaired executive functions, disinhibition, emotional lability => **frontal lobe**
- Amnesia, aggression, receptive aphasia, upper quadrantic visual field defect, temporal lobe epilepsy => **temporal lobe**
- Apraxia**: inability to perform a skilled motor activity despite intact strength, sensation, attention, memory & drive, with no other movement disorders => **left inferior parietal lobule, frontal lobes** (especially premotor cortex, supplementary motor area), or **corpus callosum**

## Localization: cerebral cortex

- Agnosia (tactile, visual), acalculia, hemineglect (non-dominant), lower quadrantic visual field defect => **parietal lobe**
- Contralateral hemianopia with macular sparing, visual hallucination => **occipital lobe**
- Disorientation & reduced consciousness, low intelligence => **global or diffuse**



## Localization: system

- **Pyramidal system:** corticospinal tract & corticobulbar tracts; direct control of motor neurons in brainstem cranial motor nuclei & spinal anterior horn; weakness pattern, spasticity
- **Extrapyramidal system:** caudate nucleus, putamen, globus pallidus, substantia nigra; modulation & indirect control of motor neurons; bradykinesia, stiffness, tremor, abnormal movements, abnormal posturing

## Localization: system

- **Cerebellar system:** coordination & rhythm of movements; unsteadiness, clumsiness, vertigo, shaky vision, scanning speech; truncal ataxia (vermis)
- **Autonomic system:** orthostatic hypotension, sexual dysfunction, decreased ability to sweat, supine hypertension, changes in gastrointestinal & urinary habits

# TELL US WHAT

JC Whole Class Session

# YOU THINK

# THANK YOU!



Look for today's teacher  
and the others you may have missed.  
(HKU Portal login required)